

Degly Sebastian Pava Pava

✉ deglypavapava@gmail.com ☎ +1 (630)635-5612 🌐 in/deglypava 🌐 www.deglypava.com

WORK EXPERIENCE

Software Engineer XR Operations

Omni Applications LLC - LucidDreams S.A.S

April 2021 - Present, United States - Colombia

- In charge of providing Extended Reality solutions across diverse sectors, including Virtual Reality tours, TikTok, Snapchat and Instagram social media filters, game development, and machine learning implementations.
- Developed filters and AR experiences that have caused more than 291.4 mil lifetime interactions according to Meta Spark Hub.
- Provided consultation to clients has resulted in a notable 20-40% surge in customer engagement.
- UX design and collaborative version control is prioritized by using Dependency Injection and Reactive Programming while communicating actively through Slack and Git to provide maintainable code and good documentation.

Unity 3D Developer

The University of British Columbia

December 2018 - July 2021, Vancouver, Canada

- Delivered impactful sessions at the BC Tech Summit, engaging around 2000+ industry partners in the assessment of Augmented Reality experiences, including the notable project "The HoloBrain" in collaboration with the Emerging Media Lab and Microsoft Garage.
- Successfully brought 3 projects to user-facing production within THE HIVE Laboratory, while overseeing their integration into a Virtual Reality environment using the Unity 3D Engine, VIVE and Oculus devices and various JavaScript frameworks.
- Implemented advanced 3D graphics and VR / AR / MR toolkits to develop seamless and interactive Mixed Reality experiences. Used by almost 100 neuroanatomy students on UBC courses and soon going global.

Research Frontend Developer

Purdue University

August 2018 - December 2018, Indiana, United States

- Proposed a robust software solution anchored in HTML, CSS, and React, strong front-end engineering principles, customized for the unique requirements of civil engineers and architects. The software used D3.js library to create elegant data visualizations.

EDUCATION

ERASMUS MUNDUS JAPAN - MEng, Lighting and XR | MSc in Optics, Image, Computer Vision and Multimedia | Msc of Computer Science

Toyohashi University of Technology | Universite Jean-Monnet | University of Eastern Finland • Japan | France | Finland • 2023 • Recipient of 8 notable academic excellence distinctions and awards. Part of the program's Quality Board.

Systems and Computer Engineering

Minor in Software Engineering • Universidad Nacional de Colombia • 2019 • Best High School Graduates Full Scholarship

PROJECTS

MRI-Derived Brain Volumetric Rendering on HoloLens

Microsoft Garage - The University of British Columbia • December 2018 - July 2021

- Engineered a C++ shader-based methodology to transform Blender created MRI brain scans into volumetric 3D visualizations on the HoloLens, seamlessly integrating principles of 3d Math, Linear Algebra, Computer Graphics, and GLSL. Unity used for UX design.

MR. Manikin, Vuforia-Enabled Mobile Clinical Simulation

The University of British Columbia • January 2020 - December 2020

- Employed augmented reality to give life to Annie, the mannequin that would teach students how to perform PCR and other medical techniques. Made use of Flutter for mobile development, Photon for synchronous multiplayer, Firebase as a database, and harnessed Slack, Jira and Git for the game design phase, facilitating effective collaboration among a team of 10+ individuals.

CERTIFICATIONS & SKILLS

Certifications: Meta Augmented Reality Developer Professional Certificate | Meta Spark Creator AR Certification | Unity and C# Proficiency | AR games using Vuforia SDK | User Experience & Interaction Design for AR/VR/MR/XR | AR (Augmented Reality) & Video Codec Streaming Services Emerging Technologies

Languages: English, French, Japanese, Spanish